# African Climate Trend Analysis

## Executive summary

Recent analyses show that Africa is warming faster than the global average, with significant increases in both land and ocean temperatures over the last three decades. Rising temperatures, increasingly erratic rainfall, and more frequent extremes (droughts, floods, and heat waves) are undermining food, water, and energy security, as well as public health and infrastructure. This report synthesizes observed climate trends, projected changes, key risks, and priority adaptation and early-warning actions for the continent.

## 1. Background and objectives

Africa's climate-sensitive sectors agriculture, water, energy, health, and coastal infrastructure are highly exposed to variability and long-term change. The objective of this analysis is to:

- summarize observed temperature and precipitation trends across Africa;
- assess recent climate extremes and their impacts;
- highlight regional differences and emerging risks; and
- Propose priority adaptation and monitoring measures.

## 2. Observed temperature trends

### 2.1 Continental-scale warming

Africa has experienced one of the most rapid rates of temperature increase on the planet, especially since the 1990s.

The World Meteorological Organization's State of the Climate in Africa reports significant area-averaged temperature increases in all six African sub regions over 1991–2024, with recent years among the warmest on record.

### 2.2 Regional and local warming

East Africa, including the Horn (e.g., Ethiopia), has shown marked warming in both minimum and maximum temperatures, with linear trends often exceeding 0.04–0.05 °C per decade in some urban stations.

Similar warning signals are documented in West and Southern Africa, where hot extremes and heat waves have become more frequent and intense.

## 3. Precipitation and hydro-climate trends

### 3.1 Rainfall variability

Continent-wide, annual rainfall shows high spatial and temporal variability, but key subregions display clear trends:

- Sahel and parts of East Africa: increased intensity of wet-season extremes, coexisting with more recurrent drought years.
- Southern Africa: rising frequency of dry years and prolonged droughts, affecting water availability.

### 3.2 Impacts on water and ecosystems

- Changes in seasonal rainfall timing and intensity are altering river flows, groundwater recharge, and wetland systems.
- These shifts contribute to water scarcity, soil degradation, and stress on forest and savannah ecosystems, reducing resilience to additional shocks.

## 4. Climate extremes and recent events

### 4.1 Extreme weather and climate events

In 2022, over 110 million people in Africa were directly affected by weather, climate, and water-related hazards, resulting in more than 8.5 billion USD in economic damages.

Droughts and floods together accounted for the majority of fatalities and displacement, with drought-related deaths alone making up nearly half of reported climate-related mortality.

### 4.2 The 2024 context

2024 ranked among the warmest years on record for Africa, with record-high sea-surface temperatures and widespread marine heat waves affecting coastal fisheries and livelihoods.

Simultaneously, extreme rainfall and flooding events in East Africa and parts of West Africa disrupted agriculture, displaced populations, and strained health systems.

5. Regional-scale differences

The table below summarises major climate-trend patterns across key sub regions:

| **Region** | **Temperature trend (recent decades)** | **Rainfall / hydro-climate trend** | **Key observed risks** |
|---|---|---|---|
| Horn of Africa (e.g., Ethiopia) | Strong warming in min/max temps; urban stations show >0.04 °C/decade. academia+1 | Increased variability with more intense drought and flood episodes. int-res+1 | Food insecurity, pastoralist livelihoods, water stress. wmo+1 |

| | | | |
|---|---|---|---|
| Sahel and West Africa | Marked warming and heat-wave increase. int-res+1 | More intense wet-season rains, but also prolonged dry spells. int-res+1 | Crop failure, displacement, energy-water stress. climate-chance |
| Southern Africa | Rising temperatures and heatwaves. wmo+1 | Increasing frequency of multi-year droughts. wmo | Water shortages, hydropower disruption, biodiversity loss. digitallibrary.un |
| East Africa (coastal) | Warming sea-surface temperatures and marine heatwaves. wmo | Rising coastal flooding and storm-related impacts. wmo | Coastal erosion, fisheries, infrastructure damage. wmo |

## 6. Future climate projections

### 6.1 Temperature and extremes

Model-based projections for Africa through the 21st century indicate continued warming under all greenhouse-gas scenarios, with higher-end emissions leading to more severe heat stress and heat waves.

Frequency and intensity of hot days are expected to increase, especially in semi-arid and arid regions, with implications for labor productivity, health, and cooling demand.

### 6.2 Precipitation and water

Simulations suggest a general increase in rainfall intensity but greater uncertainty in total annual rainfall, implying more frequent heavy precipitation events interspersed with dry periods.

This pattern is likely to increase flood risks while also exacerbating water-scarcity stress in regions already facing low-reliability rainfall.

## 7. Impacts and vulnerabilities

### 7.1 Agriculture, food, and water security

- Rising temperatures and erratic rainfall reduce crop yields and shorten growing seasons in many regions, particularly rain-fed systems.
- Water scarcity and competition for shared river basins (e.g., Nile, Niger, Zambezi) are likely to intensify, especially under lower-flow scenarios.

### 7.2 Health and livelihoods

- Warming and extreme weather events increase prevalence of heat-related illness, vector-borne diseases, and malnutrition linked to food and water insecurity.
- Informal-sector workers, pastoralists, smallholder farmers, and coastal communities are among the most vulnerable groups.

## 8. Adaptation, early warning, and monitoring

### 8.1 Strengthening early-warning systems

- Investment in hydro meteorological infrastructure, real-time data sharing, and AI-enhanced forecasting can improve lead times for floods, droughts, and heat waves.
- Mobile-based alert systems and community-led dissemination channels are critical for reaching rural and marginalised populations.

### 8.2 Priority adaptation measures

Key priority actions include:

- Climate-smart agriculture and drought-resilient cropping systems.
- Integrated water-resources management and investment in water-storage and efficiency.
- Urban planning and coastal protection to reduce flood and heat-island risks.
- Strengthening social-protection systems to buffer climate-related livelihood shocks.

### 9. Policy recommendations

- Mainstream climate-trend analysis into national-level development planning, including infrastructure, energy, and financial-sector strategies.
- Scale up investment in climate-data infrastructure, including meteorological stations, satellite data integration, and open-data platforms.
- Promote regional cooperation on shared climate risks (drought, floods, trans boundary basins) and harmonization of early-warning protocols.

10. Conclusion

Africa's climate system is undergoing rapid change, with clearly rising temperatures and increasingly erratic rainfall patterns across most subregions. These trends translate into heightened risks for food, water, energy, health, and economic stability, particularly for vulnerable populations and rain-fed economies. Building robust early-warning systems, improving climate-data availability, and investing in context-specific adaptation measures are essential to enhance resilience and sustainable development across the continent.